HW2: Reproducing Serena Williams's Plot

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Introduction

For this homework, I reproduced a scatter plot about Serena Williams's professional tennis matches. This assignment is a continuation of HW1, where I planned how to recreate the graph. In HW2, I used real match data from Tennis Abstract instead of the example data I used before.

Data and Methods

I downloaded Serena Williams's career match data from Tennis Abstract. The original data was stored in a JavaScript file, so I used R to clean and organize the information.

I calculated Serena's age and her opponent's age at the time of each match. Then I calculated the age difference by subtracting the opponent's age from Serena's age. I also separated the match results into wins and losses.

After cleaning the data, I used ggplot2 in R to create the scatter plot.

Reproduced Plot

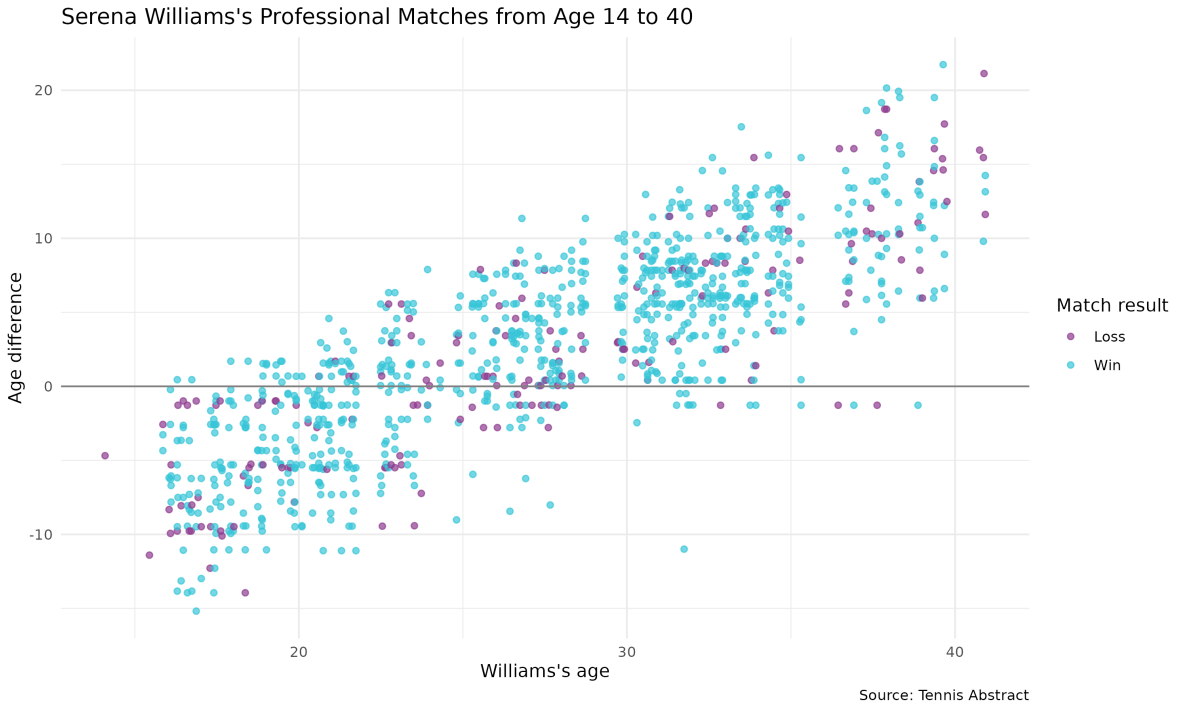


Figure 1: Serena Williams's Professional Matches

Results

The x-axis shows Serena Williams's age, and the y-axis shows the age difference between Serena and her opponent. Each dot represents one match. Blue dots represent wins, and purple dots represent losses.

When a point is above zero, Serena was older than her opponent. When a point is below zero, she was younger.

From the graph, I noticed that when Serena was younger, she mostly played against older opponents. As she got older, she played against more younger opponents. I also noticed that she won many matches throughout her career.

Conclusion

Through this assignment, I learned how to download raw data, clean it in R, and use the cleaned data to reproduce a graph. I also learned why it is important to organize files and keep the raw data separate from the cleaned data so that other people can reproduce the same results.

Data Source

Tennis Abstract: https://www.tennisabstract.com/cgi-bin/wplayer-classic.cgi?p=SerenaWilliams&f=ACareerqc